

SUPPLEMENTARY MATERIALS: Aligning Network Equivariance with Data Symmetry: A Theoretical Framework and Adaptive Approach for Image Restoration*

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SM1. Proof of Lemma 3.1.

Proof. For continuous image functions $\hat{r}(x), r(x) \in C^\infty(\mathbb{R}^2)$, we have:

$$\begin{aligned} \varphi[\pi_A[\hat{r}], \pi_A[r]] &= \arg \min_k \int_{\mathbb{R}^2} \left(\pi_A[\hat{r}](x) - \int_{\mathbb{R}^2} k(t) \pi_A[r](x-t) dt \right)^2 dx \\ &= \arg \min_k \int_{\mathbb{R}^2} \left(\hat{r}(A^{-1}x) - \int_{\mathbb{R}^2} k(t) r(A^{-1}x - A^{-1}t) dt \right)^2 dx. \end{aligned} \quad (\text{SM1.1})$$

Let $\hat{k}(t) = \pi_A^{-1}[k](t) = k(At)$, and $\hat{t} = A^{-1}t, \hat{x} = A^{-1}x$. Since $|\det(A)| = 1$, then we have:

$$\begin{aligned} \varphi[\pi_A[\hat{r}], \pi_A[r]] &= \arg \min_k \int_{\mathbb{R}^2} \left(\hat{r}(\hat{x}) - \int_{\mathbb{R}^2} k(At) r(\hat{x} - \hat{t}) d\hat{t} \right)^2 d\hat{x} \\ &= \pi_A \arg \min_{\hat{k}} \int_{\mathbb{R}^2} \left(\hat{r}(\hat{x}) - \int_{\mathbb{R}^2} \hat{k}(\hat{t}) r(\hat{x} - \hat{t}) d\hat{t} \right)^2 d\hat{x} \\ &= \pi_A[\varphi[\hat{r}, r]]. \end{aligned} \quad (\text{SM1.2})$$

The proof is completed. ■

SM2. Proof of Lemma 3.2.

Proof. According to the definition of L_2 norm, we can deduce:

$$\|\pi_A[r](x)\|_2^2 = \int_{\mathbb{R}^2} (\pi_A[r](x))^2 dx = \int_{\mathbb{R}^2} (r(A^{-1}x))^2 dx. \quad (\text{SM2.1})$$

Let $\bar{x} = A^{-1}x$, then we have

$$\|\pi_A[r](x)\|_2^2 = \int_{\mathbb{R}^2} (r(\bar{x}))^2 d\bar{x} = \|r(x)\|_2^2. \quad (\text{SM2.2}) \quad \text{■}$$

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SM3. Proof of Lemma 3.4. Before proving Lemma 3.4, we state several necessary lemmas, continuing from the framework established by Xie et al. [SM4].

Lemma SM3.1. For a smooth function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, if the following results are satisfied:

$$(SM3.1) \quad \max \left\{ \left| \frac{\partial^2 f(x)}{\partial x_1^2} \right|, \left| \frac{\partial^2 f(x)}{\partial x_2^2} \right| \right\} \leq C, \quad \forall \|x\| \geq (p+1/2)h, \quad f(x) = 0,$$

where $p, h > 0$, then we have:

$$(SM3.2) \quad \max \left\{ \left| \frac{d^2}{dx_2^2} \int_{\mathbb{R}} f(x) dx_1 \right|, \left| \frac{d^2}{dx_1^2} \int_{\mathbb{R}} f(x) dx_2 \right| \right\} \leq C(p+1)h.$$

Lemma SM3.2. For any $f : \mathbb{R} \rightarrow \mathbb{R}$, let l denote the linear interpolation function between points $x = a$ and $x = b$, i.e.,

$$(SM3.3) \quad l(x) = \frac{(x-a)f(b) + (b-x)f(a)}{b-a}.$$

If the second derivative f'' is continuous in $[a, b]$ and $|f''(x)| \leq C$, then it holds that:

$$(SM3.4) \quad |f(x) - l(x)| \leq \frac{C}{8}(a-b)^2.$$

Lemma SM3.3. For smooth functions $r : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $k : \mathbb{R}^2 \rightarrow \mathbb{R}$, if for $x \in \mathbb{R}^2$, the following conditions are satisfied:

$$(SM3.5) \quad \begin{aligned} |r(x)| &\leq F_1, \quad |k(x)| \leq F_2, \\ \|\nabla r(x)\| &\leq G_1, \quad \|\nabla k(x)\| \leq G_2, \\ \|\nabla^2 r(x)\| &\leq H_1, \quad \|\nabla^2 k(x)\| \leq H_2, \\ \forall \|x\| &\geq (p+1/2)h, \quad k(x) = 0 \end{aligned}$$

where $p, h > 0$, ∇ and ∇^2 denote the operators of gradient and Hessian matrix, respectively, then, $\forall y \in \mathbb{R}$, the following results are satisfied:

$$(SM3.6) \quad \left| \int_{\mathbb{R}^2} k(x) r(y-x) d\sigma(x) - \sum_{i,j \in \Delta} k(x_{ij}) r(y-x_{ij}) h^2 \right| \leq \frac{(p+1)^2 C}{4} h^4,$$

where $\Delta = \{(i, j) | i, j = 1, 2, \dots, p\}$, $x_{ij} = ((i - (p+1)/2)h, (j - (p+1)/2)h)^T$ and $C = F_1 H_2 + F_2 H_1 + 2G_1 G_2$.

The proofs of the above lemmas can be found in [SM4]. Now let us prove Lemma 3.4 in the main text.

Proof. 1) We first prove that for any $n \in \mathbb{Z}$, it holds that:

$$(SM3.7) \quad \left| \int_{\mathbb{R}} f \left(\begin{bmatrix} x_1 \\ nh \end{bmatrix} \right) dx_1 - \sum_{m \in \mathbb{Z}} f \left(\begin{bmatrix} mh \\ nh \end{bmatrix} \right) h \right| \leq \frac{CD}{4} h^2.$$

40 By Lemma SM3.2, we can obtain that $\forall m, n \in \mathbb{Z}, mh \leq x_1 < (m+1)h$, it holds that:

$$41 \quad (\text{SM3.8}) \quad \left| f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) - l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right| \leq \frac{C}{8}h^2,$$

42 where l_{mn} is the linear interpolation function of f between points $x = \left[\begin{smallmatrix} mh \\ nh \end{smallmatrix}\right]$ and $x =$
 43 $\left[\begin{smallmatrix} mh+h \\ nh \end{smallmatrix}\right]$, i.e.,

$$44 \quad (\text{SM3.9}) \quad l_{mn}(x) = \frac{(x_1 - mh)f\left(\left[\begin{smallmatrix} mh+h \\ nh \end{smallmatrix}\right]\right) + (mh + h - x_1)f\left(\left[\begin{smallmatrix} mh \\ nh \end{smallmatrix}\right]\right)}{h}.$$

45 Thus,

$$\begin{aligned} & \left| \int_{mh}^{mh+h} \left(f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 - \int_{mh}^{mh+h} \left(l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 \right| \\ &= \left| \int_{mh}^{mh+h} \left(f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) - l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 \right| \\ 46 \quad (\text{SM3.10}) \quad & \leq \int_{mh}^{mh+h} \left| f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) - l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right| dx_1 \\ & \leq \int_{mh}^{mh+h} \frac{C}{8}h^2 dx_1 = \frac{C}{8}h^3. \end{aligned}$$

47 Besides, we have:

$$\begin{aligned} & \left| \int_{\mathbb{R}} f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) dx_1 - \sum_{m \in \mathbb{Z}} f\left(\left[\begin{smallmatrix} mh \\ nh \end{smallmatrix}\right]\right) h \right| \\ &= \left| \int_{\mathbb{R}} f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) dx_1 - \sum_{m \in \mathbb{Z}} \frac{h}{2} \left(f\left(\left[\begin{smallmatrix} mh \\ nh \end{smallmatrix}\right]\right) + f\left(\left[\begin{smallmatrix} mh+h \\ nh \end{smallmatrix}\right]\right) \right) \right| \\ 48 \quad (\text{SM3.11}) \quad &= \left| \int_{\mathbb{R}} f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) dx_1 - \sum_{m \in \mathbb{Z}} \int_{mh}^{mh+h} \left(l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 \right| \\ & \leq \sum_{m \in \mathbb{Z}} \left| \int_{mh}^{mh+h} f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) dx_1 - \int_{mh}^{mh+h} \left(l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 \right|. \end{aligned}$$

49 Since $\forall \|x\| \geq D$, $f(x) = 0$, there are at most $2(D/h)$ values of m , s.t.,

$$50 \quad (\text{SM3.12}) \quad \left| \int_{mh}^{mh+h} \left(f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 - \int_{mh}^{mh+h} \left(l_{mn}\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) \right) dx_1 \right| \neq 0,$$

51 while all of them satisfy (SM3.10). Then, by (SM3.10), (SM3.11) and (SM3.12), we can easily achieve
 52 (SM3.7).

53 2) We then prove

$$54 \quad (\text{SM3.13}) \quad \left| \int_{\mathbb{R}^2} f(x) d\sigma(x) - \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\left[\begin{smallmatrix} x_1 \\ nh \end{smallmatrix}\right]\right) dx_1 h \right| \leq \frac{D^2 C}{2} h^2.$$

55 Let $F(x_2) = \int_{\mathbb{R}} f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) dx_1$, and then

56 (SM3.14)
$$\int_{\mathbb{R}^2} f(x) d\sigma(x) = \int_{\mathbb{R}} F(x_2) dx_2.$$

57 By Lemma SM3.1, we can obtain that $|F''(x_2)| \leq 2DC$. Let $L_n(x_2)$ be the linear interpolation function
 58 of $F(x_2)$, between points $x_2 = nh$ and $x_2 = nh + h$, and then for any $x_2 \in [nh, nh + h]$, by Lemma
 59 SM3.2, we have:

60 (SM3.15)
$$|F(x_2) - L_n(x_2)| \leq \frac{DC}{4}h^2.$$

61 Thus, similar to (SM3.10), we can deduce that:

62 (SM3.16)
$$\left| \int_{nh+h}^{nh} F(x_2) dx_2 - \int_{nh+h}^{nh} L_n(x_2) dx_2 \right| \leq \int_{nh+h}^{nh} |F(x_2) - L_n(x_2)| dx_2 \leq \frac{DC}{4}h^3.$$

63 Then, similar to (SM3.11), we can deduce that:

64 (SM3.17)
$$\begin{aligned} & \left| \int_{\mathbb{R}^2} f(x) d\sigma(x) - \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\begin{bmatrix} x_1 \\ nh \end{bmatrix}\right) dx_1 h \right| \\ &= \left| \int_{\mathbb{R}} F(x_2) dx_2 - \sum_{n \in \mathbb{Z}} F(nh) h \right| \\ &= \left| \int_{\mathbb{R}} F(x_2) dx_2 - \sum_{n \in \mathbb{Z}} \frac{1}{2}(F(nh) + F(nh + h)) h \right| \\ &\leq \sum_{n \in \mathbb{Z}} \left| \int_{nh}^{nh+h} F(x_2) dx_2 - \frac{1}{2}(F(nh) + F(nh + h)) h \right|. \end{aligned}$$

65 Since $\forall \|x\| \geq D$, $f(x) = 0$, there are at most $2(D/h)$ values of m , s.t.,

66 (SM3.18)
$$\left| \int_{nh}^{nh+h} F(x_2) dx_2 - \frac{1}{2}(F(nh) + F(nh + h)) h \right| \neq 0,$$

67 while all of them satisfy (SM3.16). Then, by (SM3.16), (SM3.17) and (SM3.18), we can easily achieve
 68 (SM3.13).

69 3) By (SM3.7) and (SM3.13), we can obtain:

70 (SM3.19)
$$\begin{aligned} & \left| \int_{\mathbb{R}^2} f(x) d\sigma(x) - \sum_{i,j \in \Lambda} f(x_{ij}) h^2 \right| \\ &= \left| \int_{\mathbb{R}^2} f(x) \sigma(x) - \sum_{m,n \in \mathbb{Z}} f\left(\begin{bmatrix} mh \\ nh \end{bmatrix}\right) h^2 \right| \\ &\leq \left| \int_{\mathbb{R}^2} f(x) \sigma(x) - \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\begin{bmatrix} x_1 \\ nh \end{bmatrix}\right) dx_1 h \right| + \sum_{n \in \mathbb{Z}} \left| \int_{\mathbb{R}} f\left(\begin{bmatrix} x_1 \\ nh \end{bmatrix}\right) dx_1 - \sum_{m \in \mathbb{Z}} f\left(\begin{bmatrix} mh \\ nh \end{bmatrix}\right) h \right| h, \end{aligned}$$

71 where, by (SM3.13),

$$72 \quad (\text{SM3.20}) \quad \left| \int_{\mathbb{R}^2} f(x) \sigma(x) - \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f \left(\begin{bmatrix} x_1 \\ nh \end{bmatrix} \right) dx_1 h \right| \leq \frac{D^2 C}{2} h^2.$$

73 Besides, Since $\forall \|x\| \geq D$, $f(x) = 0$, we only have $2(D/h)$ values of m , s.t.,

$$74 \quad (\text{SM3.21}) \quad \left| \int_{\mathbb{R}} f \left(\begin{bmatrix} x_1 \\ nh \end{bmatrix} \right) dx_1 - \sum_{m \in \mathbb{Z}} f \left(\begin{bmatrix} mh \\ nh \end{bmatrix} \right) h \right| \neq 0.$$

75 Thus, by (SM3.7), we have:

$$76 \quad (\text{SM3.22}) \quad \sum_{n \in \mathbb{Z}} \left| \int_{\mathbb{R}} f \left(\begin{bmatrix} x_1 \\ nh \end{bmatrix} \right) dx_1 - \sum_{m \in \mathbb{Z}} f \left(\begin{bmatrix} mh \\ nh \end{bmatrix} \right) h \right| h \leq \frac{D^2 C}{2} h^2.$$

77 Combining (SM3.19), (SM3.20), (SM3.22), the proof is completed. ■

78 **SM4. Proof of Lemma 3.5.**

79 *Proof.* Let $f(x) = k^R(x)r(x-y)$, then we have:

$$80 \quad (\text{SM4.1}) \quad \nabla^2 f(x) = \nabla^2 k^R(x)r(x-y) + \nabla k^R(x)\nabla r(x-y) + \nabla k^R(x)\nabla r(x-y) + k^R(x)\nabla^2 r(x-y).$$

81 Then from the known boundedness conditions:

$$82 \quad (\text{SM4.2}) \quad \begin{aligned} \|r(x)\| &\leq F_r, \quad \|k^R(x)\| \leq F_k, \\ \|\nabla r(x)\| &\leq G_r, \quad \|\nabla k^R(x)\| \leq G_k, \\ \|\nabla^2 r(x)\| &\leq H_r, \quad \|\nabla^2 k^R(x)\| \leq H_k, \end{aligned}$$

83 we can deduce that:

$$84 \quad (\text{SM4.3}) \quad \|\nabla^2 f(x)\| \leq H_k F_r + 2G_r G_k + F_k H_r.$$

85 By Lemma 3.4 (Discretization Error) in the main text, we can get:

$$86 \quad (\text{SM4.4}) \quad \left| \int_{\mathbb{R}^2} k^R(x)r(x-y)dx - \sum_{(i,j) \in \Delta} k^R(x_{ij})r(x_{ij}-y)h^2 \right| \leq (H_k F_r + 2G_r G_k + F_k H_r)(D_r + D_R)^2 h^2.$$

87 We can split $|R - \tilde{R}|$ as:

(SM4.5)

$$\begin{aligned}
 |R - \tilde{R}| &= \frac{1}{(D_r + D_R)^2} \left| \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} k^R(x) r(x - y) dx dy - \sum_{(i,j) \in \Delta} \sum_{(\hat{i}, \hat{j}) \in \Delta} k^R(x_{ij}) r(x_{i-\hat{i}, j-\hat{j}}) h^4 \right| \\
 &\leq \frac{1}{(D_r + D_R)^2} \left| \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} k^R(x) r(x - y) dx dy - \int_{\mathbb{R}^2} \left(\sum_{(i,j) \in \Delta} k^R(x_{ij}) r(x_{i-\hat{i}, j-\hat{j}}) h^2 \right) dy \right| + \\
 88 \quad &\frac{1}{(D_r + D_R)^2} \left| \int_{\mathbb{R}^2} \left(\sum_{(i,j) \in \Delta} k^R(x_{ij}) r(x_{i-\hat{i}, j-\hat{j}}) h^2 \right) dy - \sum_{(i,j) \in \Delta} \sum_{(\hat{i}, \hat{j}) \in \Delta} k^R(x_{ij}) r(x_{i-\hat{i}, j-\hat{j}}) h^4 \right| \\
 &\leq \frac{1}{(D_r + D_R)^2} \underbrace{\left| \int_{\mathbb{R}^2} \left(\int_{\mathbb{R}^2} k^R(x) r(x - y) dx - \sum_{(i,j) \in \Delta} k^R(x_{ij}) r(x_{ij} - y) h^2 \right) dy \right|}_{(1)} + \\
 &\quad \underbrace{\frac{1}{(D_r + D_R)^2} \left| \int_{\mathbb{R}^2} \left(\sum_{(i,j) \in \Delta} k^R(x_{ij}) r(x_{ij} - y) h^2 \right) dy - \sum_{(i,j) \in \Delta} \sum_{(\hat{i}, \hat{j}) \in \Delta} k^R(x_{ij}) r(x_{i-\hat{i}, j-\hat{j}}) h^4 \right|}_{(2)}
 \end{aligned}$$

89 For the part (1), using (SM4.4) and the support of y (whose area measure is bounded by
 90 $(D_r + D_R)^2$), we have:

$$91 \quad (SM4.6) \quad (1) \leq \int_{\mathbb{R}^2} (H_k F_r + 2G_r G_k + F_k H_r) (D_r + D_R)^2 h^2 dy \leq (H_k F_r + 2G_r G_k + F_k H_r) (D_r + D_R)^4 h^2.$$

92 For the part (2), based on the linearity of finite summation and integration, we can
 93 interchange the order of integration and summation:

$$94 \quad (SM4.7) \quad \int_{\mathbb{R}^2} \left(\sum_{(i,j) \in \Delta} k^R(x_{ij}) r(x_{ij} - y) h^2 \right) dy = \sum_{(i,j) \in \Delta} \left(k^R(x_{ij}) h^2 \int_{\mathbb{R}^2} r(x_{ij} - y) dy \right).$$

95 Then, substituting into (2), we obtain:

$$\begin{aligned}
 96 \quad (SM4.8) \quad (2) &= \left| \sum_{(i,j) \in \Delta} k^R(x_{ij}) h^2 \left(\int_{\mathbb{R}^2} r(x_{ij} - y) dy - \sum_{(\hat{i}, \hat{j}) \in \Delta} r(x_{ij} - y_{\hat{i}\hat{j}}) h^2 \right) \right| \\
 &\leq \sum_{(i,j) \in \Delta} |k^R(x_{ij})| h^2 \left| \int_{\mathbb{R}^2} r(x_{ij} - y) dy - \sum_{(\hat{i}, \hat{j}) \in \Delta} r(x_{ij} - y_{\hat{i}\hat{j}}) h^2 \right|.
 \end{aligned}$$

97 By Lemma 3.4 (Discretization Error) in the main text and the boundedness condition, we can
 98 deduce:

$$99 \quad (\text{SM4.9}) \quad (2) \leq \sum_{(i,j) \in \Delta} F_k h^2 (H_r(D_r + D_R)^2 h^2) = F_k H_r (D_r + D_R)^2 h^2 \left(\sum_{(i,j) \in \Delta} h^2 \right),$$

100 where $\sum_{(i,j) \in \Delta} h^2$ represents the total area of the integration domain, thus we have:

$$101 \quad (\text{SM4.10}) \quad (2) \leq F_k H_r (D_r + D_R)^4 h^2.$$

102 Combining the bound of (1) and (2), we can get:

$$103 \quad (\text{SM4.11}) \quad \begin{aligned} |R - \tilde{R}| &\leq \frac{1}{(D_r + D_R)^2} ((H_k F_r + 2G_r G_k + 2F_k H_r)(D_r + D_R)^4 h^2) \\ &= (H_k F_r + 2G_r G_k + 2F_k H_r)(D_r + D_R)^2 h^2. \end{aligned}$$

104 The proof is completed. ■

105 **SM5. Proof of Theorem 3.6.**

106 **SM5.1. The Analysis of Semi-Discrete Operator .** Before diving into the fully discrete
 107 formulation, we introduce a *semi-discrete operator*, denoted as $\bar{\Phi}[\hat{r}_n]$ (defined in (SM5.1)),
 108 which minimizes the discretized problem (approximated via Riemann sums) but searches for
 109 the optimal solution within the continuous function space Ω_ε . The equivariance property
 110 of the *semi-discrete operator* $\bar{\Phi}[\hat{r}_n]$ evaluated in the L_2 function space is summarized as the
 111 following Lemma.

112 **Lemma SM5.1.** *For the dataset $\{\hat{r}_n\}_{n=1}^N$ and the following semi-discrete operator:*
 (SM5.1)

$$113 \quad \{\bar{\Phi}[\hat{r}_n]\}_{n=1}^N = \arg \min_{\{r_n\}_{n=1}^N \in \Omega_\varepsilon} \sum_{n=1}^N \left(\min_k \sum_{(i,j) \in \Delta} \left(\hat{r}_n(x_{ij}) - \sum_{(u,v) \in \Gamma} k(t_{uv}) r_n(x_{ij} - t_{uv}) h^2 \right)^2 h^2 \right) + \tilde{R}[r_n],$$

114 where $\Delta = \{(i,j) | i, j = 1, 2, \dots, m\}$, $m = 2D_r/h + 1$, $\Gamma = \{(u,v) | u, v = 1, 2, \dots, l\}$, $l = 2D_k/h + 1$,
 115 and Ω_ε is the relaxed symmetric feasible set. Suppose the objective function satisfies the μ -
 116 quadratic growth condition in a local neighborhood of the optimal solution (as in Theorem 3.3
 117 in the main text), and the following boundedness conditions hold for the underlying image
 118 functions $\hat{r}(x), r(x)$, the blur kernel $k(x)$, and the kernel function $k^R(x)$ associated with the
 119 regularizer R :

(SM5.2)

$$\begin{aligned} 120 \quad &|\hat{r}(x)|, |r(x)| \leq F_1, \quad |k(x)| \leq F_2, \quad |k^R(x)| \leq F_3 \\ &\|\nabla \hat{r}(x)\|, \|\nabla r(x)\| \leq G_1, \quad \|\nabla k(x)\| \leq G_2, \quad \|\nabla k^R(x)\| \leq G_3, \\ &\|\nabla^2 \hat{r}(x)\|, \|\nabla^2 r(x)\| \leq H_1, \quad \|\nabla^2 k(x)\| \leq H_2, \quad \|\nabla^2 k^R(x)\| \leq H_3, \\ &\forall \|x\| \geq D_r, \quad \hat{r}(x) = 0 \text{ and } r(x) = 0, \quad \forall \|x\| \geq D_k, \quad k(x) = 0, \quad \forall \|x\| \geq D_R, \quad k^R(x) = 0, \end{aligned}$$

where ∇ and ∇^2 denotes the operators of gradient and Hessian matrix, respectively. Then, for any transformation $A_n \in G_n, n=1, 2, \dots, N$ with $|\det(A_n)| = 1$, we have:

$$(SM5.3) \quad \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \leq \bar{C}h^2 + \hat{C}\varepsilon,$$

where $\bar{C} = \frac{4}{\mu} [C_{\text{reg}}(D_r + D_R)^2 + \bar{C}(D_r + D_k)^2]$, $C_{\text{reg}} = H_3F_1 + 2G_1G_3 + F_3H_1$, $\bar{C} = 2(1 + 4D_k^2F_2)[4C_0D_k^2F_1 + (G_1^2 + F_1H_1)(1 + 4D_k^2F_2)]$, $C_0 = F_1H_2 + F_2H_1 + 2G_1G_2$, and $\hat{C} = \frac{2}{\mu}$.

Before proving Lemma SM5.1, we first present the following necessary lemmas and corresponding proofs.

Lemma SM5.2. For function $G(x) = \hat{r}(x) - \int_{\mathbb{R}^2} k(t)r(x-t)dt$, we define:

$$(SM5.4) \quad I = \int_{\mathbb{R}^2} (G(x))^2 dx = \int_{\mathbb{R}^2} \left(\hat{r}(x) - \int_{\mathbb{R}^2} k(t)r(x-t)dt \right)^2 dx,$$

and its discretization form:

$$(SM5.5) \quad \tilde{I} = \sum_{i,j \in \Delta} \left(\tilde{G}(x_{ij}) \right)^2 h^2 = \sum_{i,j \in \Delta} \left(\hat{r}(x_{ij}) - \sum_{u,v \in \Gamma} k(t_{uv})r(x_{ij} - t_{uv}) \right)^2 h^2,$$

where $\Delta = \{(i, j) | i, j = 1, 2, \dots, m\}$, $m = 2D_r/h + 1$, $\Gamma = \{(u, v) | u, v = 1, 2, \dots, l\}$, $l = 2D_k/h + 1$. If the following conditions are satisfied:

$$(SM5.6) \quad \begin{aligned} &|\hat{r}(x)|, |r(x)| \leq F_1, \quad |k(x)| \leq F_2, \\ &\|\nabla \hat{r}(x)\|, \|\nabla r(x)\| \leq G_1, \quad \|\nabla k(x)\| \leq G_2, \\ &\|\nabla^2 \hat{r}(x)\|, \|\nabla^2 r(x)\| \leq H_1, \quad \|\nabla^2 k(x)\| \leq H_2, \\ &\forall \|x\| \geq D_r, \quad \hat{r}(x) = 0 \text{ and } r(x) = 0, \quad \forall \|x\| \geq D_k, \quad k(x) = 0, \end{aligned}$$

where ∇ and ∇^2 denote the operators of gradient and Hessian matrix, respectively. Then the following results is satisfied:

$$(SM5.7) \quad |I - \tilde{I}| \leq \bar{C}(D_r + D_k)^2 h^2.$$

where $\bar{C} = 2(1 + 4D_k^2F_2)[4C_0D_k^2F_1 + (G_1^2 + F_1H_1)(1 + 4D_k^2F_2)]$, $C_0 = F_1H_2 + F_2H_1 + 2G_1G_2$.

Proof. We have $G(x) = \hat{r}(x) - \int_{\mathbb{R}^2} k(t)r(x-t)dt$, and $\tilde{G}(x) = \hat{r}(x) - \sum_{u,v \in \Gamma} k(t_{uv})r(x - t_{uv})h^2$. For the subsequent proofs, we first deduce the upper bounds for $G(x)$, $\tilde{G}(x)$, $\nabla \tilde{G}(x)$, and $\nabla^2 \tilde{G}(x)$.

$$(SM5.8) \quad |G(x)| = \left| \hat{r}(x) - \int_{\mathbb{R}^2} k(t)r(x-t)dt \right| \leq |\hat{r}(x)| + \int_{\mathbb{R}^2} |k(t)| |r(x-t)| dt \leq F_1 + 4D_k^2F_1F_2.$$

143 Similarly,

(SM5.9)

$$144 \quad \left| \tilde{G}(x) \right| = \left| \hat{r}(x) - \sum_{u,v \in \Gamma} k(t_{uv}) r(x - t_{uv}) h^2 \right| \leq |\hat{r}(x)| + \sum_{u,v \in \Gamma} |k(t_{uv})| |r(x - t_{uv})| h^2 \leq F_1 + 4D_k^2 F_1 F_2.$$

145 For the gradient and Hessian matrix of $\tilde{G}(x)$, we can deduce:

$$146 \quad \begin{aligned} \left| \nabla \tilde{G}(x) \right| &= \left| \nabla \hat{r}(x) - \sum_{u,v \in \Gamma} k(t_{uv}) \nabla r(x - t_{uv}) h^2 \right| \\ &\leq |\nabla \hat{r}(x)| + \sum_{u,v \in \Gamma} |k(t_{uv})| |\nabla r(x - t_{uv})| h^2 \\ &\leq G_1 + 4D_k^2 F_2 G_1. \end{aligned}$$

147

$$148 \quad \begin{aligned} \left| \nabla^2 \tilde{G}(x) \right| &= \left| \nabla^2 \hat{r}(x) - \sum_{u,v \in \Gamma} k(t_{uv}) \nabla^2 r(x - t_{uv}) h^2 \right| \\ &\leq |\nabla^2 \hat{r}(x)| + \sum_{u,v \in \Gamma} |k(t_{uv})| |\nabla^2 r(x - t_{uv})| h^2 \\ &\leq H_1 + 4D_k^2 F_2 H_1. \end{aligned}$$

149 For $I = \int_{\mathbb{R}^2} (G(x))^2 dx$, and its discretization form $\tilde{I} = \sum_{i,j \in \Delta} \left(\tilde{G}(x_{ij}) \right)^2 h^2$, we can split (SM5.7)
 150 as:

(SM5.12)

$$151 \quad \left| I - \tilde{I} \right| \leq \underbrace{\left| \int_{\mathbb{R}^2} (G(x))^2 dx - \int_{\mathbb{R}^2} \left(\tilde{G}(x) \right)^2 dx \right|}_{\langle 1 \rangle} + \underbrace{\left| \int_{\mathbb{R}^2} \left(\tilde{G}(x) \right)^2 dx - \sum_{i,j \in \Delta} \left(\tilde{G}(x_{ij}) \right)^2 h^2 \right|}_{\langle 2 \rangle},$$

152 For the part $\langle 1 \rangle$ of Eq. (SM5.12), we have:

$$153 \quad \left(\text{SM5.13} \right) \quad \left| \int_{\mathbb{R}^2} (G(x))^2 dx - \int_{\mathbb{R}^2} \left(\tilde{G}(x) \right)^2 dx \right| = \left| \int_{\mathbb{R}^2} \left(G(x) - \tilde{G}(x) \right) \left(G(x) + \tilde{G}(x) \right) dx \right|.$$

154 By Lemma SM3.3, we know that:

$$155 \quad \left(\text{SM5.14} \right) \quad \left| \int_{\mathbb{R}^2} k(t) r(x - t) dt - \sum_{u,v \in \Gamma} k(t_{uv}) r(x - t_{uv}) h^2 \right| \leq D_k^2 C_0 h^2,$$

156 where $C_0 = F_1 H_2 + F_2 H_1 + 2G_1 G_2$. Then we have:

(SM5.15)

$$157 \quad \left| G(x) - \tilde{G}(x) \right| = \left| \hat{r}(x) - \int_{\mathbb{R}^2} k(t) r(x-t) dt - \hat{r}(x) + \sum_{u,v \in \Gamma} k(t_{uv}) r(x-t_{uv}) h^2 \right| \leq D_k^2 C_0 h^2.$$

158 According to (SM5.8) and (SM5.9), we know that $|G(x)| \leq F_1 + 4D_k^2 F_1 F_2$, $|\tilde{G}|(x) \leq F_1 +$
 159 $4D_k^2 F_1 F_2$, so that $|G(x) + \tilde{G}(x)| \leq 2F_1 + 8D_k^2 F_1 F_2$. Then, we can deduce:

$$160 \quad \begin{aligned} \text{(SM5.16)} \quad \left| \int_{\mathbb{R}^2} (G(x))^2 dx - \int_{\mathbb{R}^2} (\tilde{G}(x))^2 dx \right| &\leq D_k^2 C_0 h^2 (2F_1 + 8D_k^2 F_1 F_2) 4(D_r + D_k)^2. \\ &= 4 [2C_0 D_k^2 F_1 (1 + 4D_k^2 F_2)] (D_r + D_k)^2 h^2. \end{aligned}$$

161 For the part (2) of (SM5.12), it corresponds to the discretization error of $\int_{\mathbb{R}^2} (\tilde{G}(x))^2 dx$. If
 162 the second-order partial derivatives of $(\tilde{G}(x))^2$ are bounded, Lemma 3.4 becomes applicable.
 163 Thus We proceed with an analysis of the second-order partial derivatives. We deduce that:

$$164 \quad \begin{aligned} \text{(SM5.17)} \quad \frac{\partial(\tilde{G}(x))^2}{\partial x_1} &= 2\tilde{G}(x) \frac{\partial \tilde{G}(x)}{\partial x_1}, \quad \frac{\partial^2(\tilde{G}(x))^2}{\partial x_1^2} = 2 \left[\left(\frac{\partial \tilde{G}(x)}{\partial x_1} \right)^2 + \tilde{G}(x) \frac{\partial^2(\tilde{G}(x))}{\partial x_1^2} \right]. \\ \frac{\partial(\tilde{G}(x))^2}{\partial x_2} &= 2\tilde{G}(x) \frac{\partial \tilde{G}(x)}{\partial x_2}, \quad \frac{\partial^2(\tilde{G}(x))^2}{\partial x_2^2} = 2 \left[\left(\frac{\partial \tilde{G}(x)}{\partial x_2} \right)^2 + \tilde{G}(x) \frac{\partial^2(\tilde{G}(x))}{\partial x_2^2} \right]. \end{aligned}$$

165 Then by (SM5.10) and (SM5.11), we have:

$$166 \quad \begin{aligned} \text{(SM5.18)} \quad \max \left\{ \left| \frac{\partial^2(\tilde{G}(x))^2}{\partial x_1^2} \right|, \left| \frac{\partial^2(\tilde{G}(x))^2}{\partial x_2^2} \right| \right\} &\leq 2((G_1 + 4D_k^2 F_2 G_1)^2 + (F_1 + 4D_k^2 F_1 F_2)(H_1 + 4D_k^2 F_2 H_1)). \end{aligned}$$

167 According to Lemma 3.4, we have:

$$168 \quad \begin{aligned} \text{(SM5.19)} \quad \left| \int_{\mathbb{R}^2} (\tilde{G}(x))^2 dx - \sum_{i,j \in \Delta} (\tilde{G}(x_{ij}))^2 h^2 \right| \\ \leq 2((G_1 + 4D_k^2 F_2 G_1)^2 + (F_1 + 4D_k^2 F_1 F_2)(H_1 + 4D_k^2 F_2 H_1)) (D_r + D_k)^2 h^2. \end{aligned}$$

Thus, combining (SM5.16) and (SM5.19), we can achieve:

$$\begin{aligned}
 & (SM5.20) \\
 & \quad |I - \tilde{I}| \\
 & \leq 4(2C_0 D_k^2 F_1 (1 + 4D_k^2 F_2)) (D_r + D_k)^2 h^2 + \\
 & \quad 2((G_1 + 4D_k^2 F_2 G_1)^2 + (F_1 + 4D_k^2 F_1 F_2)(H_1 + 4D_k^2 F_2 H_1)) (D_r + D_k)^2 h^2 \\
 & = [8C_0 D_k^2 F_1 (1 + 4D_k^2 F_2) + 2(G_1^2 (1 + 4D_k^2 F_2)^2 + F_1 H_1 (1 + 4D_k^2 F_2)^2)] (D_r + D_k)^2 h^2 \\
 & = [8C_0 D_k^2 F_1 (1 + 4D_k^2 F_2) + 2(G_1^2 + F_1 H_1)(1 + 4D_k^2 F_2)^2] (D_r + D_k)^2 h^2 \\
 & = 2(1 + 4D_k^2 F_2) [4C_0 D_k^2 F_1 + (G_1^2 + F_1 H_1)(1 + 4D_k^2 F_2)] (D_r + D_k)^2 h^2 \\
 & = \bar{C} (D_r + D_k)^2 h^2,
 \end{aligned}$$

where $\bar{C} = 2(1 + 4D_k^2 F_2) [4C_0 D_k^2 F_1 + (G_1^2 + F_1 H_1)(1 + 4D_k^2 F_2)]$, $C_0 = F_1 H_2 + F_2 H_1 + 2G_1 G_2$.

The proof is completed. ■

Lemma SM5.3. For any two functions $g(k)$ and $\hat{g}(k)$, if the following condition is satisfied:

$$(SM5.21) \quad |g(k) - \hat{g}(k)| \leq C,$$

then it holds that:

$$(SM5.22) \quad \left| \min_k g(k) - \min_k \hat{g}(k) \right| \leq C.$$

Proof. For $\forall g(k), \hat{g}(k)$, we can assume that $\min_k g(k) \geq \min_k \hat{g}(k)$. And let

$$(SM5.23) \quad k^* = \arg \min_k g(k), \quad \hat{k}^* = \arg \min_k \hat{g}(k).$$

Then we have:

$$\begin{aligned}
 & (SM5.24) \\
 & \left| \min_k g(k) - \min_k \hat{g}(k) \right| = \min_k g(k) - \min_k \hat{g}(k) = g(k^*) - \hat{g}(\hat{k}^*) \leq g(\hat{k}^*) - \hat{g}(\hat{k}^*) \leq |g(\hat{k}^*) - \hat{g}(\hat{k}^*)| \leq C.
 \end{aligned}$$

Now, based on the lemmas above, we proceed to prove Lemma SM5.1.

Proof. We explicitly define the continuous objective functionals for the n -th sample as follows:

$$(SM5.25) \quad F[r_n] = \left(\min_{k \in \Lambda} \int_{\mathbb{R}^2} \left(\hat{r}_n(x) - \int_{\mathbb{R}^2} k(t) r_n(x-t) dt \right)^2 dx \right) + R[r_n],$$

which satisfies the μ -quadratic growth condition [SM1, SM2], that is, for the optimal solution r_n^* and any r_n ,

$$(SM5.26) \quad \sum_{n=1}^N F[r_n] - F[r_n^*] \geq \mu \sum_{n=1}^N \|r_n - r_n^*\|_2^2.$$

189 Then we denote the discrete form of (SM5.25) as:

$$190 \quad (\text{SM5.27}) \quad \tilde{F}[r_n] = \left(\min_{k \in \Lambda} \sum_{i,j \in \Delta} \left(\hat{r}_n(x_{ij}) - \sum_{u,v \in \Gamma} k(t_{uv}) r_n(x_{ij} - t_{uv}) h^2 \right)^2 h^2 \right) + \tilde{R}[r_n].$$

191 By Lemma 3.5 (Discretization Error of Regularization Term) in the main text,
192 we can obtain:

$$193 \quad (\text{SM5.28}) \quad |R - \tilde{R}| \leq (H_3 F_1 + 2G_1 G_3 + 2F_3 H_1)(D_r + D_R)^2 h^2,$$

194 By Lemma SM5.2, we can obtain that:

$$195 \quad (\text{SM5.29}) \quad \left| \int_{\mathbb{R}^2} \left(\hat{r}_n(x) - \int_{\mathbb{R}^2} k(t) r_n(x-t) dt \right)^2 dx - \sum_{i,j \in \Delta} \left(\hat{r}_n(x_{ij}) - \sum_{u,v \in \Gamma} k(t_{uv}) r_n(x_{ij} - t_{uv}) h^2 \right)^2 h^2 \right| \leq \bar{C}(D_r + D_k)^2 h^2.$$

196 where $\bar{C} = 2(1 + 4D_k^2 F_2) [4C_0 D_k^2 F_1 + (G_1^2 + F_1 H_1)(1 + 4D_k^2 F_2)]$, $C_0 = F_1 H_2 + F_2 H_1 + 2G_1 G_2$.

197 Then by Lemma SM5.3, we have:

$$198 \quad (\text{SM5.30}) \quad \left| \min_{k \in \Lambda} \int_{\mathbb{R}^2} \left(\hat{r}_n(x) - \int_{\mathbb{R}^2} k(t) r_n(x-t) dt \right)^2 dx - \min_{k \in \Lambda} \sum_{i,j \in \Delta} \left(\hat{r}_n(x_{ij}) - \sum_{u,v \in \Gamma} k(t_{uv}) r_n(x_{ij} - t_{uv}) h^2 \right)^2 h^2 \right| \leq \bar{C}(D_r + D_k)^2 h^2.$$

199 Thus, combining (SM5.28) and (SM5.30), we can achieve:

$$200 \quad (\text{SM5.31}) \quad |F[r_n] - \tilde{F}[r_n]| \leq (H_3 F_1 + 2G_1 G_3 + 2F_3 H_1)(D_r + D_R)^2 h^2 + \bar{C}(D_r + D_k)^2 h^2.$$

201 Then, for the restoration operator Φ and $\bar{\Phi}$:

$$202 \quad (\text{SM5.32}) \quad [\Phi(\hat{r}_n)] = \arg \min_{[r_n] \in \Omega_\varepsilon} \sum_{n=1}^N F[r_n], \quad [\bar{\Phi}(\hat{r}_n)] = \arg \min_{[r_n] \in \Omega_\varepsilon} \sum_{n=1}^N \tilde{F}[r_n],$$

203 with μ -quadratic growth condition, we can deduce:

$$\begin{aligned} \sum_{n=1}^N \|\Phi[\hat{r}_n] - \bar{\Phi}[\hat{r}_n]\|_2^2 &\leq \frac{1}{\mu} \sum_{n=1}^N F[\bar{\Phi}[\hat{r}_n]] - F[\Phi[\hat{r}_n]] \\ &= \frac{1}{\mu} \sum_{n=1}^N \left(F[\bar{\Phi}[\hat{r}_n]] - \tilde{F}[\bar{\Phi}[\hat{r}_n]] + \tilde{F}[\bar{\Phi}[\hat{r}_n]] - F[\Phi[\hat{r}_n]] \right) \\ 204 \quad (\text{SM5.33}) \quad &\leq \frac{1}{\mu} \sum_{n=1}^N \left| F[\bar{\Phi}[\hat{r}_n]] - \tilde{F}[\bar{\Phi}[\hat{r}_n]] \right| + \left| \tilde{F}[\bar{\Phi}[\hat{r}_n]] - F[\Phi[\hat{r}_n]] \right| \\ &\leq \frac{1}{\mu} \sum_{n=1}^N \left| F[\bar{\Phi}[\hat{r}_n]] - \tilde{F}[\bar{\Phi}[\hat{r}_n]] \right| + \left| \tilde{F}[\Phi[\hat{r}_n]] - F[\Phi[\hat{r}_n]] \right|. \end{aligned}$$

By substituting the uniform error bound between F and $\tilde{F}$ established in (SM5.31), we obtain:

$$(SM5.34) \quad \sum_{n=1}^N \|\Phi[\hat{r}_n] - \bar{\Phi}[\hat{r}_n]\|_2^2 \leq \frac{2N}{\mu} ((H_3F_1 + 2G_1G_3 + 2F_3H_1)(D_r + D_R)^2h^2 + \bar{C}(D_r + D_k)^2h^2).$$

Now we split the (SM5.3) as:

$$(SM5.35) \quad \begin{aligned} & \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \\ & \leq \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \Phi[\pi_{A_n}[\hat{r}_n]]\|_2^2 + \frac{1}{N} \sum_{n=1}^N \|\Phi[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\Phi[\hat{r}_n]]\|_2^2 + \\ & \quad \frac{1}{N} \sum_{n=1}^N \|\pi_{A_n}[\Phi[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2. \end{aligned}$$

With (SM5.34) we have:

$$(SM5.36) \quad \frac{1}{N} \sum_{n=1}^N \|\Phi[\pi_{A_n}[\hat{r}_n]] - \bar{\Phi}[\pi_{A_n}[\hat{r}_n]]\|_2^2 \leq \frac{2}{\mu} ((H_3F_1 + 2G_1G_3 + 2F_3H_1)(D_r + D_R)^2h^2 + \bar{C}(D_r + D_k)^2h^2).$$

By **Theorem 3.3 (Equivariance Error of Restoration Operator)** in the main text, we can get:

$$(SM5.37) \quad \frac{1}{N} \sum_{n=1}^N \|\Phi[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\Phi[\hat{r}_n]]\|_2^2 \leq \frac{2\varepsilon}{\mu}.$$

From (SM5.34) and Lemma 3.2 in the main text, which states that the transformation π_{A_n} preserves the L_2 norm, we obtain:

$$(SM5.38) \quad \frac{1}{N} \sum_{n=1}^N \|\pi_{A_n}[\Phi[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \leq \frac{2}{\mu} ((H_3F_1 + 2G_1G_3 + 2F_3H_1)(D_r + D_R)^2h^2 + \bar{C}(D_r + D_k)^2h^2).$$

Finally, by combining (SM5.36), (SM5.37), and (SM5.38), we can get:

$$(SM5.39) \quad \begin{aligned} & \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \\ & \leq \frac{4}{\mu} ((H_3F_1 + 2G_1G_3 + 2F_3H_1)(D_r + D_R)^2h^2 + \bar{C}(D_r + D_k)^2h^2) + \frac{2\varepsilon}{\mu} \\ & = \tilde{C}h^2 + \hat{C}\varepsilon, \end{aligned}$$

where $\tilde{C} = \frac{4}{\mu} [C_{\text{reg}}(D_r + D_R)^2 + \bar{C}(D_r + D_k)^2]$, $C_{\text{reg}} = H_3 F_1 + 2G_1 G_3 + 2F_3 H_1$, $\bar{C} = 2(1 + 4D_k^2 F_2) [4C_0 D_k^2 F_1 + (G_1^2 + F_1 H_1)(1 + 4D_k^2 F_2)]$, $C_0 = F_1 H_2 + F_2 H_1 + 2G_1 G_2$, and $\hat{C} = \frac{2}{\mu}$.

The proof is completed. ■

SM5.2. Proof of Theorem 3.6 in the main text.

Proof. Based on the boundedness condition $|\hat{r}(x)|, |r(x)| \leq F_1$ stated in Theorem 3.6 in the main text, we have:

$$(SM5.40) \quad |\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]| \leq |\bar{\Phi}[\pi_{A_n}[\hat{r}_n]]| + |\pi_{A_n}[\bar{\Phi}[\hat{r}_n]]| \leq 2F_1.$$

To streamline the subsequent proof, we define:

$$(SM5.41) \quad E_n = \bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]].$$

Then we have $|E_n| \leq 2F_1$. For $\|\nabla E_n\|$, we can deduce:

$$(SM5.42) \quad \nabla E_n(x) = \nabla \bar{\Phi}[\pi_{A_n}[\hat{r}_n]](x) - \nabla \pi_{A_n}[\bar{\Phi}[\hat{r}_n]](x) = \nabla \bar{\Phi}[\pi_{A_n}[\hat{r}_n]](x) - (A_n^{-1})^T \nabla [\bar{\Phi}[\hat{r}_n]](A_n^{-1}x).$$

Then,

$$(SM5.43) \quad \|\nabla E_n(x)\| \leq \|\nabla \bar{\Phi}[\pi_{A_n}[\hat{r}_n]](x)\| + \|(A_n^{-1})^T\| \|\nabla [\bar{\Phi}[\hat{r}_n]](A_n^{-1}x)\| \leq 2G_1.$$

Similarly, for $\|\nabla^2 E_n\|$, we have:

$$(SM5.44) \quad \|\nabla^2 E_n(x)\| \leq \|\nabla^2 \bar{\Phi}[\pi_{A_n}[\hat{r}_n]](x)\| + \|(A_n^{-1})^T\| \|\nabla^2 [\bar{\Phi}[\hat{r}_n]](A_n^{-1}x)\| \|A_n^{-1}\| \leq 2H_1.$$

Based on the above results, we now proceed to analyze the second-order partial derivatives of $(E_n)^2$, from which it follows:

$$(SM5.45) \quad \begin{aligned} \frac{\partial(E_n)^2}{\partial x_1} &= 2E_n \frac{\partial E_n}{\partial x_1}, & \frac{\partial^2(E_n)^2}{\partial x_1^2} &= 2 \left[\left(\frac{\partial E_n}{\partial x_1} \right)^2 + E_n \frac{\partial^2(E_n)}{\partial x_1^2} \right]. \\ \frac{\partial(E_n)^2}{\partial x_2} &= 2E_n \frac{\partial E_n}{\partial x_2}, & \frac{\partial^2(E_n)^2}{\partial x_2^2} &= 2 \left[\left(\frac{\partial E_n}{\partial x_2} \right)^2 + E_n \frac{\partial^2(E_n)}{\partial x_2^2} \right]. \end{aligned}$$

Then combine (SM5.40), (SM5.43), and (SM5.44), we have:

$$(SM5.46) \quad \max \left\{ \left| \frac{\partial^2(E_n)^2}{\partial x_1^2} \right|, \left| \frac{\partial^2(E_n)^2}{\partial x_2^2} \right| \right\} \leq 2(4G_1^2 + 4F_1 H_1) = 8G_1^2 + 8F_1 H_1.$$

Given the boundedness condition in (SM5.46), **Lemma 3.4 (Discretization Error)** implies:

$$(SM5.47) \quad \left| \int_{\mathbb{R}^2} (E_n(x))^2 d\sigma(x) - \sum_{i,j \in \Delta} (E_n(x_{ij}))^2 h^2 \right| \leq (8G_1^2 + 8F_1 H_1)(D_r + D_k)^2 h^2.$$

In other words, we can obtain the following conclusion:

(SM5.48)

$$\left| \int_{\mathbb{R}^2} (\bar{\Phi}[\pi_{A_n}[\hat{r}_n(x)]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n(x)]])^2 dx - \sum_{i,j \in \Delta} (\bar{\Phi}[\pi_{A_n}[\hat{r}_n(x_{ij})]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n(x_{ij})]])^2 h^2 \right| \leq (8G_1^2 + 8F_1H_1)(D_r + D_k)^2 h^2.$$

The target we aim to analyze is the operator $\tilde{\Phi}(\hat{Y}_n)$ in the discrete domain. For $\bar{\Phi}[\hat{r}_n]$, its L_2 norm is given by $\|\bar{\Phi}[\hat{r}_n]\|_2^2 = \int_{\mathbb{R}^2} (\bar{\Phi}[\hat{r}_n](x))^2 dx$. And for $\tilde{\Phi}(\hat{Y}_n)$, the corresponding Frobenius norm is given by $\|\tilde{\Phi}(\hat{Y}_n)\|_F^2 = \sum_{i,j \in \Delta} (\tilde{\Phi}[\hat{r}_n](x_{ij}))^2 h^2$, where $\Delta = \{(i, j) | i, j = 1, 2, \dots, m\}$, $m = 2D/h + 1$, h is the mesh size. Then (SM5.48) is equivalent to the following:

$$\left| \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 - \frac{1}{N} \sum_{n=1}^N \|\tilde{\Phi}(\tilde{\pi}_{A_n}(\hat{Y}_n)) - \tilde{\pi}_{A_n}(\tilde{\Phi}(\hat{Y}_n))\|_F^2 \right| \leq (8G_1^2 + 8F_1H_1)(D_r + D_k)^2 h^2.$$

Now, we split the target to be proved as follows:

(SM5.50)

$$\frac{1}{N} \sum_{n=1}^N \|\tilde{\Phi}(\tilde{\pi}_{A_n}(\hat{Y}_n)) - \tilde{\pi}_{A_n}(\tilde{\Phi}(\hat{Y}_n))\|_F^2 = \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 + \left(\frac{1}{N} \sum_{n=1}^N \|\tilde{\Phi}(\tilde{\pi}_{A_n}(\hat{Y}_n)) - \tilde{\pi}_{A_n}(\tilde{\Phi}(\hat{Y}_n))\|_F^2 - \frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \right).$$

By Lemma SM5.1, we have obtained:

$$\frac{1}{N} \sum_{n=1}^N \|\bar{\Phi}[\pi_{A_n}[\hat{r}_n]] - \pi_{A_n}[\bar{\Phi}[\hat{r}_n]]\|_2^2 \leq \tilde{C}h^2 + \hat{C}\varepsilon.$$

Therefore, by combining (SM5.49) and (SM5.51), we can deduce:

$$\frac{1}{N} \sum_{n=1}^N \|\tilde{\Phi}(\tilde{\pi}_{A_n}(\hat{Y}_n)) - \tilde{\pi}_{A_n}(\tilde{\Phi}(\hat{Y}_n))\|_F^2 \leq \tilde{C}h^2 + \hat{C}\varepsilon + (8G_1^2 + 8F_1H_1)(D_r + D_k)^2 h^2 = [\tilde{C} + (8G_1^2 + 8F_1H_1)(D_r + D_k)^2]h^2 + \hat{C}\varepsilon.$$

where $\tilde{C} = \frac{4}{\mu} [C_{\text{reg}}(D_r + D_R)^2 + \bar{C}(D_r + D_k)^2]$, $C_{\text{reg}} = H_3F_1 + 2G_1G_3 + 2F_3H_1$, $\bar{C} = 2(1 + 4D_k^2F_2) [4C_0D_k^2F_1 + (G_1^2 + F_1H_1)(1 + 4D_k^2F_2)]$, $C_0 = F_1H_2 + F_2H_1 + 2G_1G_2$, and $\hat{C} = \frac{2}{\mu}$.

Let $C_1 = \frac{4}{\mu}(H_3F_1 + 2G_1G_3 + 2F_3H_1)$, $C_2 = \frac{4}{\mu}\bar{C} + 8G_1^2 + 8F_1H_1$, then the total error term can be simplified and combined as $[C_1(D_r + D_R)^2 + C_2(D_r + D_k)^2]h^2 + \hat{C}\varepsilon$.

The proof is completed. ■

261 **SM6. Proof of Lemma 4.1.** Before proving Lemma 4.1, we first present the following
 262 lemma and its proof.

263 **Lemma SM6.1.** *For a transformation group G , π_A represents how the transformation ma-*
 264 *trix $A \in G$ acts on a image function $r(x)$, i.e., $\pi_A[r](x) = r(A^{-1}x), \forall x \in \mathbb{R}^2$. Then for any*
 265 *image Y , where $Y_{ij} = r(x_{ij})$, we have $\pi_A[\pi_{\bar{A}}](Y) = \pi_{A\bar{A}}(Y)$,*

266 *Proof.* We can deduce that:

$$267 \quad (\text{SM6.1}) \quad (\pi_A[\pi_{\bar{A}}](Y))_{ij} = \pi_A[\pi_{\bar{A}}[r]](x_{ij}) = r(\bar{A}^{-1}A^{-1}x_{ij}) = r((A\bar{A})^{-1}x_{ij}) = (\pi_{A\bar{A}}(Y))_{ij}. \blacksquare$$

268 Based on Lemma SM6.1 above, we now present the proof of Lemma 4.1 in the main text.

269 *Proof.* For any image Y , by Lemma SM6.1, we have:

$$\begin{aligned} 270 \quad (\text{SM6.2}) \quad \bar{\Psi}(\pi_{\bar{A}}(Y)) &= \sum_{A \in G} \frac{1}{|G|} \pi_A(\Psi(\pi_{\bar{A}}^{-1}\pi_{\bar{A}}))(Y) \\ &= \sum_{A \in G} \frac{1}{|G|} \pi_{\bar{A}}\pi_{\bar{A}^{-1}}\pi_A(\Psi(\pi_{A^{-1}\bar{A}}))(Y) \\ &= \pi_{\bar{A}} \left(\sum_{A \in G} \frac{1}{|G|} \pi_{\bar{A}^{-1}A}(\Psi(\pi_{\bar{A}^{-1}A}^{-1})) \right) (Y) \\ &= \pi_{\bar{A}}(\bar{\Psi}(Y)). \end{aligned}$$

271 The proof is completed. ■

272 **SM7. Transformation Adaptive Equivariant Convolutions.**

273 **SM7.1. The framework in Continuous Domain.** The convolution for input layer is de-
 274 noted as Ψ_w , with w being the adaptive transformation parameters. It maps an input
 275 $r \in C^\infty(\mathbb{R}^2)$ to a feature map defined on $E(2)$. Specifically, $\forall(x, A) \in E(2)$:

$$276 \quad (\text{SM7.1}) \quad \Psi_w[r](x, A) = \int_{\mathbb{R}^2} \varphi_{in}(A^{-1}D_w^{-1}\delta)r(x - \delta)d\sigma(\delta),$$

277 where σ is the measure on $\mathbb{R}^2$, A denotes the rotation transformation, which is also the index
 278 of group dimension, D_w is the adaptive transformation, and φ_{in} is the parameterized filter as
 279 defined in the main text.

280 Let Φ_w denote the convolution in the intermediate layer, which maps a feature map
 281 $e \in C^\infty(E(2))$ to another, then for $\forall(x, B) \in E(2)$:

$$282 \quad (\text{SM7.2}) \quad \Phi_w[e](x, B) = \int_{O(2)} \int_{\mathbb{R}^2} \varphi_A(B^{-1}D_w^{-1}\delta)e(x - \delta, BA)d\sigma(\delta)dv(A),$$

283 where v is a measure on $O(2)$ ¹, $A, B \in O(2)$ denote transformations in the considered group,
 284 and φ_A is the parameterized filter.

¹ $O(2)$ is the orthogonal group.

Let Υ_w denote the convolution in the output layer, which maps a feature map $e \in C^\infty(E(2))$ to a function defined on $\mathbb{R}^2$, for $\forall x \in \mathbb{R}^2$:

$$(SM7.3) \quad \Upsilon_w[e](x) = \int_{O(2)} \int_{\mathbb{R}^2} \varphi_{out}(B^{-1}D_w^{-1}\delta) e(x - \delta, B) d\sigma(\delta) dv(B),$$

where $B \in O(2)$ and φ_{out} is the parameterized filter.

SM7.2. The framework in Discrete Domain. The filters in (5.3) in the main text are in the continuous domain, while in practical applications, we can only use their discrete form. The discretization of the continuous formulation can be easily performed following the strategy of previous work [SM3, SM4].

Specifically, for $i, j = 1, 2, \dots, p$, $A, B \in S^2$ and transformation D_w , we define the discrete filters for the input layer as:

$$(SM7.4) \quad (\tilde{\Psi}_w)_{ij}^A = \sum_k v_k^{in} \varphi_k(A^{-1}D_w^{-1}\delta_{ij}) := \varphi_{in}(A^{-1}D_w^{-1}\delta_{ij}),$$

where $\delta_{ij} = ((i - (p+1)/2)h, (j - (p+1)/2)h)^T$ is the fixed mesh grid, φ_{in} is the parameterized filter of input layer, and v_k^{in} is its coefficients.

Similarly, we define the discrete filters for intermediate and output layers as follows:

$$(SM7.5) \quad (\tilde{\Phi}_w)_{ij}^{B,A} = \varphi_A(B^{-1}D_w^{-1}\delta_{ij}),$$

$$(SM7.6) \quad (\tilde{\Upsilon}_w)_{ij}^B = \varphi_{out}(B^{-1}D_w^{-1}\delta_{ij}),$$

where φ_A indicates the parameterized filter with respect to the channel of feature map indexed by A in intermediate layer, and φ_{out} is the parameterized filter in the output layer.

Based on the continuous convolutions and the discrete filters defined above, we can construct the discrete convolutions.

We assume that an image $Y \in \mathbb{R}^{n \times n}$ represents a two-dimensional grid function obtained by discretizing a smooth function, i.e., for $i, j = 1, 2, \dots, n$,

$$(SM7.7) \quad Y_{ij} = r(x_{ij}),$$

where $x_{ij} = ((i - \frac{n+1}{2})h, (j - \frac{n+1}{2})h)^T$ and h is the mesh size. F represents a three-dimensional grid function sampled from a smooth function $e : \mathbb{R}^2 \times S \rightarrow \mathbb{R}$, i.e., for $i, j = 1, 2, \dots, n$,

$$(SM7.8) \quad F_{ij}^A = e(x_{ij}, A),$$

where $A \in S$. Accordingly, we can discretize the continuous convolutions for the input layer, intermediate layer, and output layer (i.e., (SM7.1), (SM7.2), and (SM7.3)). For $\forall A \in S, \forall w \in$

²The rotation transformation group S is defined as: $S = \left\{ A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \mid \theta = 2\pi k/t, k = 0, 1, \dots, t-1 \right\}$.

315 $\mathbb{R}^3$ and $i, j = 1, 2, \dots, n$, $\tilde{\Psi}_w$ denotes the convolution of input layer, which can be calculated
 316 by:

$$317 \quad (\text{SM7.9}) \quad \left(\tilde{\Psi}_w \star Y \right)^A = (\tilde{\Psi}_w)^A * Y, \quad \forall A \in S.$$

318 Similarly, for any $B \in S$ and $i, j = 1, 2, \dots, n$, we can define the discrete convolutions of
 319 intermediate layer as the discretization of (SM7.2):

$$320 \quad (\text{SM7.10}) \quad \left(\tilde{\Phi}_w \star F \right)^B = \sum_{A \in S} \left(\tilde{\Phi}_w \right)^{B,A} * F^{BA}, \quad \forall B \in S.$$

321 To ensure efficient implementation, we avoid channel transformations on feature maps and
 322 instead use an equivalent form of the above equation:

$$323 \quad (\text{SM7.11}) \quad \left(\tilde{\Phi}_w \star F \right)^B = \sum_{A \in S} \left(\tilde{\Phi}_w \right)^{B, B^{-1}A} * F^A, \quad \forall B \in S.$$

324 It is clear that (SM7.11) can be implemented by applying the commonly used convolution on
 325 F and a $p \times p \times t \times t$ filter rotated spatially and shifted cyclically along the group dimension.

326 Then the convolution of output layer can be formulated as:

$$327 \quad (\text{SM7.12}) \quad \tilde{\Upsilon}_w \star F = \sum_{B \in S} \left(\tilde{\Upsilon}_w \right)^B * F^B, \quad \forall A \in S.$$

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